

02.3 Tool Codes and Domains

Coding Protocol: Tool Code Definitions and Domain Assignment Rules

This document defines the tool codes used to classify responsible gambling (RG) interventions across the 410 extraction records in the review. Codes are organised into 8 sections that partition the tool space by mechanism of action. Each code definition includes the defining features of the tool (inclusion criteria), the neighbouring codes from which the tool must be distinguished (exclusion criteria), and an exemplar record.

For the 34 records (8.3% of the dataset) coded with bundled combinations (e.g., FEED-PERSONAL + MSG-POP), the bundled-code assignment rule is given in the final section of this protocol. The 14 lower-level domain prefixes used in some analyses are derived from the first component of each tool code and are distinct from the 8 sections used to organise this protocol; the correspondence is given in Table 02.3.1.

Table 02.3.1
Correspondence between Code-Prefix Domains and Sections

Code prefix	Codes in prefix	Section
MSG	Messaging and Communications (MSG-POP, MSG-STATIC, MSG-NOTIFY, MSG-PUSH)	Messaging interventions
LIMIT	Limit setting (LIMIT-DEP, LIMIT-LOSS, LIMIT-TIME, LIMIT-BET, LIMIT-WIN)	Financial consumption limits
EXCL	Exclusion and Account Controls (EXCL-LONG, EXCL-TEMP)	Exclusion and account controls
FEED	Personalised Feedback (FEED-PERSONAL, FEED-NORM, FEED-RISK, FEED-TRACK)	Feedback and tracking
THER	Therapeutic Interventions (THER-CBT, THER-MI, THER-SELF, THER-COMBO)	Therapeutic interventions
EDU	Education and Information (EDU-INFO, EDU-INTERACT, EDU-VIDEO)	Education and self-assessment
ASSESS	Self-Assessment Tools (ASSESS-SELF)	Education and self-assessment
CONTACT	Proactive Contact (CONTACT-PROACTIVE)	Proactive contact, detection, and self-regulation
DETECT	Detection and Monitoring Systems (DETECT)	Proactive contact, detection, and self-regulation
BREAK	Mandatory breaks (BREAK-FORCE)	Proactive contact, detection, and self-regulation
SELF-REG	Self-Regulation Strategies (SELF-REG)	Proactive contact, detection, and self-regulation
GUIDE	Low-Risk Gambling Guidelines (GUIDE-LRGL)	Guidelines, structural modifications, and multi-component programmes
STRUCT	Structural Game Modifications (STRUCT-TEMPO, STRUCT-STAKE, STRUCT-PAY, STRUCT-DISPLAY, STRUCT-OTHER)	Guidelines, structural modifications, and multi-component programmes
PROGRAM	Multi-Component Programmes and System Tools (PROGRAM)	Guidelines, structural modifications, and multi-component programmes

Note. The code prefixes correspond to the leading segment of each Tool_Code code. The sections organise codes by broader mechanism of action and are used for structural purposes in this protocol; the taxonomy domains are retained for quantitative cross-tabulation (e.g., domain × implementation readiness tables in the main manuscript).

2. Tool Code Definitions

Each definition below follows a four-part structure: the formal definition of the tool, the features that establish inclusion under the code, the neighbouring codes from which the tool must be distinguished, and an exemplar record from the dataset. Counts (n) refer to records coded with the base code alone; records using bundled combinations are assigned per the rules in the bundled-code section below.

2.1 Messaging Interventions

MSG-POP: Pop-up messages (n = 33)

Definition. Session-triggered automated messages appearing during play that pause gambling activity and require player acknowledgement or dismissal before play resumes.

Inclusion. Interruptive messages that suspend play; content may be generic safer-gambling prompts, self-appraisal questions, or odds-information reminders.

Exclusion. Non-interruptive triggered alerts permitting continued play (→ MSG-NOTIFY); always-visible labels or signage (→ MSG-STATIC); messages delivered outside the gambling session (→ MSG-PUSH); pop-ups containing personalised data as the key mechanism (→ FEED-PERSONAL + MSG-POP bundled code).

Example. Pop-up corrective randomness message during play (Cloutier 2006); forced 15-second pause pop-up (Monaghan 2010).

MSG-STATIC: Static warning messages and labels (n = 30)

Definition. Permanent, always-visible warning text, labels, or venue signage displayed continuously without interaction requirement.

Inclusion. Passive-exposure fixed content: EGM warning stickers, venue signage, on-screen odds information, standardised safer-gambling slogans (e.g., “When the Fun Stops, Stop”), landing-page text on gambling websites.

Exclusion. Triggered alerts or dynamic periodic prompts (→ MSG-NOTIFY); interruptive pop-ups (→ MSG-POP); messages in marketing or educational brochures (→ EDU-INFO where the primary purpose is knowledge transfer).

Example. Static government-mandated warning sign on EGM frame (Monaghan 2007); “When the Fun Stops, Stop” campaign slogan (Newall 2020).

MSG-NOTIFY: Triggered in-session notifications (n = 7)

Definition. Dynamic alerts triggered by play conditions or time that appear during play without interrupting it; play continues uninterrupted while the message is visible.

Inclusion. Scrolling banners, threshold-triggered peripheral displays, animated periodic prompts, translucent on-screen messages that surface and then fade without requiring response.

Exclusion. Messages that pause play (→ MSG-POP); permanent signage (→ MSG-STATIC); out-of-session delivery (→ MSG-PUSH).

Example. Dynamic translucent scrolling warning message on screen (Monaghan 2007); animated responsible-gambling periodic on-screen prompts (Monaghan 2009).

MSG-PUSH: Push notifications (n = 4)

Definition. Operator-initiated automated messages delivered to the player account or device outside of active play via SMS, email, or app push notification.

Inclusion. Post-session summaries, threshold-crossing alerts delivered outside the session, SMS behaviour-change programmes, withdrawal reminders, loss-notification push messages.

Exclusion. One-to-one personalised outreach from the operator (→ CONTACT-PROACTIVE, which is typically triggered by risk detection and involves human contact); in-session messages of any kind.

Example. Personalised winning-streak withdrawal nudge (Auer 2024); SMS behaviour-change support programme (Rodda 2018).

2.2 Financial Consumption Limits

LIMIT-DEP: Deposit limits ($n = 19$)

Definition. Player-set or operator-default caps on amounts deposited into the gambling account over a defined time period (daily, weekly, monthly).

Inclusion. Voluntary deposit limit tools; operator-default limits applied at registration; cross-operator deposit caps.

Exclusion. Limits on amount wagered per bet (→ LIMIT-BET); limits on net losses (→ LIMIT-LOSS); limits on time spent (→ LIMIT-TIME).

Example. €1,000 daily deposit cap with voluntary lower options (Auer 2021); monthly self-imposed deposit cap (Broda 2008).

LIMIT-LOSS: Loss limits ($n = 7$)

Definition. Player-set caps on net losses over a defined time period.

Inclusion. Session-level or longer-period net-loss thresholds; hard-lock vs soft-limit variants; mandatory loss caps imposed by regulation.

Exclusion. Caps on deposits (→ LIMIT-DEP) or wagers (→ LIMIT-BET).

Example. Self-imposed monetary loss limits with session expenditure cap (Heirene 2021); hard vs soft loss limit comparison (Wohl 2024).

LIMIT-TIME: Time and session limits ($n = 6$)

Definition. Player-set caps on session duration or cumulative login time.

Inclusion. Voluntary time-limit pop-ups; self-set session duration caps; alarm-clock reminders set by the player.

Exclusion. Operator-imposed mandatory breaks (→ BREAK-FORCE).

Example. Time-limit pre-commitment pop-up with free-text session duration entry (Hing 2024); player-set alarm-clock session reminder (Wohl 2020).

LIMIT-BET: Bet and stake limits ($n = 2$)

Definition. Player-set or regulatory caps on maximum stake per wager.

Inclusion. Trial-by-trial pre-commitment stake restrictions; EGM maximum bet caps set by the player.

Exclusion. Universal regulatory EGM stake reductions applied to all players (→ STRUCT-STAKE); aggregate spending caps (→ LIMIT-DEP, LIMIT-LOSS).

Example. Trial-by-trial pre-commitment option with sequential voluntary pre-bet stake restriction (Stewart 2015).

LIMIT-WIN: Win limits ($n = 2$)

Definition. Player-set caps on wins or win-quarantine strategies encoded into platform features.

Inclusion. Bank-meter win-quarantine; self-imposed win ceilings.

Exclusion. Ad hoc self-regulatory strategies not supported by platform features (→ SELF-REG).

Example. Bank meter win quarantine feature (Blaszczynski 2014).

2.3 Exclusion and Account Controls

Note. The EXCL-* prefix here (EXCL-LONG, EXCL-TEMP) is namespace-distinct from the screening exclusion codes in 01.2 Eligibility Criteria, which use the same prefix for a different code space.

EXCL-LONG: Long-term self-exclusion ($n = 22$)

Definition. Voluntary account block for an extended period, typically six months to permanent.

Inclusion. Multi-operator exclusion schemes; operator-level permanent bans; national self-exclusion registers.

Exclusion. Short-duration voluntary blocks ($\rightarrow$ EXCL-TEMP); enforced breaks during a session ($\rightarrow$ BREAK-FORCE); parameter caps that permit continued play ($\rightarrow$ LIMIT-*).

Example. Voluntary account closure system for problem gambling (Hayer 2010); multi-operator self-exclusion register (Pickering 2018).

EXCL-TEMP: Temporary self-exclusion and cooling-off ($n = 7$)

Definition. Short-term voluntary account suspension, typically days to weeks.

Inclusion. Cooling-off periods; time-outs of hours to days; short-duration self-initiated breaks at account level.

Exclusion. Long-duration exclusions ($\rightarrow$ EXCL-LONG); session-level mandatory breaks ($\rightarrow$ BREAK-FORCE).

Example. Seven-day online gambling ban (Håkansson 2023); time-out temporary account suspension (Hopfgartner 2023).

2.4 Feedback and Tracking

FEED-PERSONAL: Personalised behavioural feedback ($n = 5$)

Definition. Tailored feedback showing a player their own gambling data (expenditure, wins/losses, session statistics) without normative comparison or risk classification.

Inclusion. Individualised displays of actual vs perceived spend; personal loss-history visualisations; win/loss estimation tools.

Exclusion. Generic content shown to all players ($\rightarrow$ MSG-*); comparison to peers or population ($\rightarrow$ FEED-NORM); classification into risk tiers ($\rightarrow$ FEED-RISK); historical activity records without interpretation ($\rightarrow$ FEED-TRACK).

Example. Personalised behavioural feedback with 6-month loss history visualisation (Wohl 2017); win/loss estimation comparison tool (Auer 2014).

FEED-NORM: Normative feedback ($n = 8$)

Definition. Feedback comparing a player's behaviour to a reference group (peers, operator average, or population).

Inclusion. Peer-comparison displays; operator-average comparisons; population-percentile statements ("you gamble more than 90% of players").

Exclusion. Population statistics presented as context without individual comparison ($\rightarrow$ EDU-INFO); risk-tier classification ($\rightarrow$ FEED-RISK).

Example. Personalised normative feedback report (Cunningham 2012); peer-loss comparison to platform average (Auer 2017).

FEED-RISK: Risk-level feedback ($n = 9$)

Definition. Feedback classifying a player's risk level presented to the player (traffic light, PGSI category, algorithmic risk score).

Inclusion. Traffic-light dashboards; PGSI-based risk tiers shown to the player; algorithmic risk scores displayed on the player's account page.

Exclusion. Operator-side algorithms that do not surface risk to the player (→ DETECT); self-administered risk screens where the player completes a questionnaire (→ ASSESS-SELF).

Example. PlayScan traffic-light behavioural feedback system (Forsström 2019); algorithmic risk dashboard (Finkenwirth 2021).

FEED-TRACK: Activity statements and tracking ($n = 7$)

Definition. Passive behavioural tracking dashboards or periodic spending statements displaying historical activity without interpretation, classification, or normative comparison.

Inclusion. Account-history downloads; monthly activity statements; transaction summaries without risk framing.

Exclusion. Personalised feedback with interpretation (→ FEED-PERSONAL); risk classification (→ FEED-RISK).

Example. Regular financial statements with activity history display (Christensen 2024); gambling expense overview download (Auer 2022).

2.5 Therapeutic Interventions

THER-CBT: Cognitive-behavioural therapy ($n = 15$)

Definition. Structured CBT delivered face-to-face, by telephone, or online, targeting cognitive distortions and gambling behaviour.

Inclusion. Manualised CBT protocols; guided online CBT programmes; cognitive restructuring interventions.

Exclusion. Self-directed workbooks without clinician contact (→ THER-SELF); brief motivational interventions (→ THER-MI); multi-modal programmes (→ THER-COMBO).

Example. Cognitive therapy for gambling disorder with erroneous belief restructuring (Ladouceur 2006); guided CBT with weekly therapist-emailed modules (Carlbring 2012).

THER-MI: Motivational interviewing ($n = 12$)

Definition. MI-based brief intervention delivered by telephone, face-to-face, or online; targets ambivalence and readiness to change.

Inclusion. Single-session MI; brief motivational enhancement therapy; telephone-based MI.

Exclusion. Structured CBT (→ THER-CBT); combined MI+CBT programmes (→ THER-COMBO); MI-flavoured proactive operator contact (→ CONTACT-PROACTIVE where the intervention is operator-initiated outreach based on risk detection).

Example. Single-session face-to-face motivational interview (Diskin 2009); brief MET session (Hodgins 2009).

THER-SELF: Self-directed therapy and online workbooks ($n = 20$)

Definition. Structured self-help programmes or workbooks delivered online or via app, typically containing CBT- or MI-based content, without direct clinician contact.

Inclusion. CBT-based workbooks; online self-help toolkits; bibliotherapy with structured content.

Exclusion. Informal player-driven strategies without structured content (→ SELF-REG); workbooks delivered alongside clinician support (→ THER-COMBO).

Example. Self-help workbook for problem gambling recovery (Hodgins 2001); online self-directed CBT programme (Cunningham 2019).

THER-COMBO: Combined therapeutic approaches ($n = 10$)

Definition. Multi-modal therapeutic programmes combining CBT, MI, and/or other clinical approaches within a single protocol.

Inclusion. MET+CBT programmes; internet-delivered CBT/MI self-help with telephone support; integrated cognitive-behavioural and desensitisation protocols.

Exclusion. Single-modality interventions ($\rightarrow$ THER-CBT, THER-MI, or THER-SELF as appropriate).

Example. MET + CBT four-session combined intervention (Toneatto 2014); internet-delivered CBT/MI self-help with telephone support (Carlbring 2008).

2.6 Education and Self-Assessment

EDU-INFO: Information-based education ($n = 11$)

Definition. Factual information about gambling odds, house edge, randomness, or problem gambling, delivered via leaflets, websites, or in-session informational displays; primary purpose is knowledge transfer rather than prompting.

Inclusion. Odds-information sheets; house-edge explainer brochures; educational website content; in-session explainer pop-ups where the primary purpose is explanation.

Exclusion. Always-visible warning labels where the primary purpose is prompting ($\rightarrow$ MSG-STATIC); interactive tutorials requiring engagement ($\rightarrow$ EDU-INTERACT); video content ($\rightarrow$ EDU-VIDEO); self-assessment tools ($\rightarrow$ ASSESS-SELF).

Example. Slot machine odds information sheet (Williams 2007); return-to-player educational materials format comparison (Newall 2022).

EDU-INTERACT: Interactive education ($n = 8$)

Definition. Interactive educational tools, quizzes, simulations, or cognitive-training tasks designed to increase gambling literacy through active engagement.

Inclusion. Digital gambling simulators for learning; analytic-thinking training with feedback; in-person facilitated education workshops.

Exclusion. Passive information display ($\rightarrow$ EDU-INFO); video content ($\rightarrow$ EDU-VIDEO); cognitive-behavioural therapy ($\rightarrow$ THER-CBT).

Example. Digital gambling accelerator with scratch-off ticket simulation (Turner 2008); analytic priming task (Hollingshead 2017).

EDU-VIDEO: Video-based education ($n = 9$)

Definition. Educational video or animation content about gambling risks or safer gambling practices.

Inclusion. Educational animations; video-based RG training modules; video components of bundled interventions when tested as a distinct mechanism.

Exclusion. Interactive tutorials with active engagement ($\rightarrow$ EDU-INTERACT); static information displays ($\rightarrow$ EDU-INFO).

Example. Educational animation on EGM function and limit setting (Wohl 2010); three-minute simplified RG animation (Wohl 2013).

ASSESS-SELF: Self-assessment tools ($n = 6$)

Definition. Player-facing screening instruments for self-identification of gambling risk level, completed by the player.

Inclusion. Self-administered PGSI; PlayScan self-diagnostic; Positive Play Scale self-assessment.

Exclusion. Operator-calculated risk classifications shown to the player (→ FEED-RISK); operator-side algorithmic risk detection (→ DETECT).

Example. PlayScan self-diagnostic test (Griffiths 2009); Positive Play Scale self-assessment (Wood 2017).

2.7 Proactive Contact, Detection, and Self-Regulation

CONTACT-PROACTIVE: Proactive operator contact (*n* = 13)

Definition. Operator-initiated one-to-one outreach to players identified as at-risk; delivered by telephone, email, or letter; MI-based content is common.

Inclusion. Telephone MI outreach; personalised letter or email outreach to flagged players; operator-triggered advisor calls.

Exclusion. Automated push notifications to broad user populations (→ MSG-PUSH); clinical MI delivered to treatment-seeking populations (→ THER-MI).

Example. Proactive telephone contact with MI-based expenditure feedback (Jonsson 2020); proactive written contact with personalised expenditure feedback letter (Rodda 2018).

DETECT: Behavioural-marker detection systems (*n* = 11)

Definition. Operator-side machine learning or algorithmic models classifying player risk from behavioural telemetry; typically without direct player-facing output.

Inclusion. Behavioural-marker risk-profiling systems; machine learning harm-prediction models; early-warning detection algorithms for operator use.

Exclusion. Risk classifications displayed to the player (→ FEED-RISK); player-administered self-assessments (→ ASSESS-SELF).

Example. Behavioural markers of harm product-risk-profiling system (Haeusler 2016); first-month behavioural marker profile for high-risk detection (LaBrie 2009).

BREAK-FORCE: Mandatory play breaks (*n* = 10)

Definition. Operator-imposed mandatory pauses of defined duration that interrupt gambling sessions, with or without player active engagement to resume.

Inclusion. Mandatory session interruptions; deposit-triggered forced breaks; regulatory minimum break durations.

Exclusion. Player-set session-duration caps (→ LIMIT-TIME); interruptive safer-gambling pop-ups that permit immediate dismissal (→ MSG-POP).

Example. Mandatory 15-minute forced session termination (Hopfgartner 2022); imposed break in play (Auer 2023).

SELF-REG: Self-regulatory strategies (*n* = 30)

Definition. Player-driven informal strategies for managing gambling behaviour, reported by players but not implemented as a platform tool: pre-session budgeting, alternative leisure activities, access avoidance, cash-only gambling, response inhibition training.

Inclusion. Strategies extracted from self-report, qualitative interviews, or ecological momentary assessment studies where the “tool” is the player’s own behaviour rather than an operator deployment.

Exclusion. Structured self-help programmes with defined content (→ THER-SELF); operator-deployed limit-setting features (→ LIMIT-*); any tool with formal operator implementation.

Example. Self-regulatory strategy of fixed spending amount (Moore 2012); alternative leisure activity strategy (Lostutter 2014).

2.8 Guidelines, Structural Modifications, and Multi-Component Programmes

GUIDE-LRGL: Low-risk gambling guidelines ($n = 13$)

Definition. Evidence-based consumption thresholds defining low-risk gambling frequency and expenditure, derived from epidemiological data; typically not yet deployed as player-facing tools.

Inclusion. Empirically-derived quantitative thresholds for low-risk play; threshold replication studies; cross-national guideline validation.

Exclusion. Guidelines implemented as operational tools that alert at threshold crossings (→ code the implementing tool, e.g., MSG-NOTIFY if a threshold-triggered alert is added).

Example. Low-risk gambling limits based on empirically-derived consumption thresholds (Currie 2006); replication of low-risk gambling limits using Canadian provincial data (Currie 2008).

STRUCT-TEMPO: Game-speed structural modifications ($n = 6$)

Definition. Modifications to reel-spin speed, inter-trial pause duration, or forced delay between bets; imposed on the product rather than set by the player.

Inclusion. EGM reel-spin speed reductions; mandatory post-outcome pauses; bet-to-outcome interval modifications.

Exclusion. Player-set session-duration caps (→ LIMIT-TIME); operator-imposed mid-session breaks (→ BREAK-FORCE).

Example. EGM reel spin speed 5-second delay (Blaszczynski 2005); forced post-outcome pause (Ladouceur 2005).

STRUCT-STAKE: Stake-size structural modifications ($n = 4$)

Definition. Regulatory or operator-imposed maximum bet or stake caps applied universally to the product, affecting all players regardless of preference.

Inclusion. EGM maximum bet reductions imposed by regulation; line-limit restrictions; universal stake caps.

Exclusion. Player-set bet limits (→ LIMIT-BET).

Example. EGM maximum bet A\$1 cap (Blaszczynski 2005); stake and line limit modifications (Thomas 2020).

STRUCT-PAY: Payment architecture modifications ($n = 6$)

Definition. Modifications to payment mechanisms, bill acceptor limits, cash-out procedures, ATM access, or payment-method restrictions at the product or venue level.

Inclusion. Bill acceptor caps; ATM access bans in gambling venues; demo-mode payout-parity regulation; cashless payment restrictions.

Exclusion. Player-set deposit limits on accounts (→ LIMIT-DEP).

Example. EGM bill acceptor A\$20 maximum (Blaszczynski 2005); cash-out realisation procedure with website transfer (Wohl 2010).

STRUCT-DISPLAY: Structural display modifications ($n = 7$)

Definition. Permanent modifications to product display elements providing feedback to all players: cumulative outcome dashboards, session timers, credit/cash toggles, built-in visualisation features.

Inclusion. Always-visible session timers; cumulative outcome dashboards built into the product; credit-to-cash balance toggles; activity trackers built into the EGM interface.

Exclusion. Personalised feedback presented as a distinct tool (→ FEED-PERSONAL); player-set toggles that are not structural product features.

Example. Cumulative outcome dashboard with real-time session visualisation (Schellinck 2014); permanent on-screen clock with continuous time display (Stewart 2014).

STRUCT-OTHER: Other structural modifications (*n* = 10)

Definition. Game-mechanics modifications not captured by TEMPO, STAKE, PAY, or DISPLAY categories: auto-play, stopping-device, skill-based features, or other structural alterations.

Inclusion. Auto-play removal or restriction; stopping-device elimination; skill-based gambling machine features; discriminatory motor-choice procedures.

Exclusion. Modifications falling under the four named STRUCT subcodes (use the named code instead).

Example. Stopping device removal eliminating illusory control (Ladouceur 2005); skill-based gambling machine with third-person gaming feature (Myrseth 2021).

PROGRAM: Multi-component operator programmes (*n* = 7)

Definition. Integrated operator-level responsible gambling programmes combining multiple tools across risk tiers, evaluated as a whole programme rather than as individual components.

Inclusion. Evaluations of GameSense, PlayScan (programme-level), Swedish monopoly RG framework; consumer-attitude assessments of bundled RG tool suites; policy audits of multi-component operator programmes.

Exclusion. Evaluations of single tools within a programme (code the specific tool); regulatory guideline assessments without a deployed programme (→ GUIDE-LRGL).

Example. GameSense integrated land-based RG programme (Wohl 2017); Finnish monopoly self-regulation framework (Selin 2019).

3. Section Definitions and Boundary Rules

The sections group the codes by mechanism of action. Within each section, subcodes refine the mechanism by content, delivery, or target. The following rules specify how tools that plausibly fit more than one section are assigned; they supplement the exclusion criteria given in each individual code definition.

3.1 Messaging Interventions vs Education and Self-Assessment

Both sections involve in-product or on-platform text content; they are distinguished by primary purpose. Messaging (MSG-*) codes capture prompting, warning, or reminding; the intended mechanism is attentional rather than knowledge-transfer. Education (EDU-*) codes capture tools designed to change a player's knowledge, beliefs, or skills; engagement is typically active (reading, watching, completing) rather than passive.

Assignment rule. If the tool's primary purpose is to prompt, warn, or notify, code MSG-*. If the primary purpose is knowledge transfer, code EDU-*. A pop-up stating "gambling odds are always against you" is MSG-POP because it is a prompt; an interactive tutorial teaching how random number generators work is EDU-INTERACT because it builds a skill. When a tool clearly does both (for example, a video shown before play that explains odds and then prompts the player), use the bundled form (e.g., EDU-VIDEO + MSG-POP). See bundled-code rules below for when bundling is warranted.

3.2 Messaging Interventions vs Feedback and Tracking

Both sections deliver in-product content to the player. They are distinguished by content personalisation. Messaging codes apply when the content is generic (the same for all players); Feedback codes apply when the content is derived from the individual player's behavioural data. The delivery vehicle (pop-up, notification, static display) does not determine the primary code: the content source does.

Assignment rule. If the content varies with the player's own gambling data, code FEED-*. If the content is identical across players, code MSG-*. When a pop-up is the delivery vehicle for personalised data and the study tests both content and delivery as mechanisms, use the bundled code FEED-PERSONAL + MSG-POP. A pop-up saying "you have gambled for 60 minutes" is personalised (FEED-PERSONAL + MSG-POP if delivered as an interruptive pop-up). A pop-up saying "set yourself a time limit" is generic (MSG-POP).

3.3 Financial Consumption Limits vs Structural Modifications

Both can cap the same underlying parameter (e.g., bet size). They are distinguished by whether the cap is player-set or universally imposed. LIMIT-* codes apply when the player (or a player-facing default) can set or adjust the cap. STRUCT-* codes apply when the cap is a product-level modification applied to all players regardless of preference (e.g., a regulatory maximum-bet reduction from A\$10 to A\$1 imposed on all EGMs in a jurisdiction).

Assignment rule. Is the parameter adjustable by the player or tied to a player-specific default? If yes, code LIMIT-*. If the parameter is imposed on all players at the product or regulatory level, code STRUCT-*. This rule also distinguishes player-set time caps (LIMIT-TIME) from operator-imposed mandatory breaks (BREAK-FORCE).

3.4 Feedback Risk Classification vs Detection Systems

Both involve operator-side behavioural data and risk assessment. They are distinguished by whether the risk rating is surfaced to the player. FEED-RISK applies when the player sees the classification. DETECT applies when the algorithm operates in the background to flag accounts for operator review without player-facing output.

Assignment rule. Does the player see their risk rating? If yes, code FEED-RISK. If the output is purely for operator use, code DETECT. If a DETECT system triggers player-facing outreach, the outreach is coded CONTACT-PROACTIVE; the detection system itself remains DETECT.

3.5 Self-Regulatory Strategies vs Therapeutic Self-Help vs Limit-Setting Features

All three can involve a player managing their own gambling behaviour. SELF-REG captures informal strategies extracted from self-report or qualitative data where the "tool" is the player's own behaviour and there is no formal operator implementation. THER-SELF captures structured self-help programmes with defined content (CBT- or MI-based workbooks, online modules, bibliotherapy). LIMIT-* captures operator-deployed platform features that enable limit-setting.

Assignment rule. Is there an operator-deployed tool or feature? If yes, code LIMIT-* (or the relevant operator-tool code). If the study describes a structured self-help programme with defined content, code THER-SELF. If the study describes informal strategies that players report using without formal tool support, code SELF-REG.

3.6 Proactive Operator Contact vs Push Notifications vs Therapeutic Motivational Interviewing

All three may involve out-of-session contact with a player, and all three may use MI-consistent language. CONTACT-PROACTIVE is defined by operator initiation based on risk detection: the contact is targeted at a flagged individual, typically one-to-one, and commonly delivered by telephone or personalised letter. MSG-PUSH is defined by automation and broad delivery: messages are sent to

user populations or triggered by generic conditions, without individualised human contact. THER-MI is defined by clinical context: the intervention is delivered to a treatment-seeking population within a clinical protocol.

Assignment rule. Is the player treatment-seeking and is the intervention delivered within a clinical protocol? If yes, THER-MI. Is the contact targeted to a flagged individual based on operator-side risk detection, with individualised content? If yes, CONTACT-PROACTIVE. Is the message automated and broadly delivered? If yes, MSG-PUSH.

3.7 Multi-Component Programmes vs Individual Tools

PROGRAM is reserved for evaluations of operator or jurisdictional RG frameworks as integrated systems: for example, GameSense, PlayScan at the programme level, or national operator audits. When a paper evaluates a single component of a programme (e.g., the limit-setting module within GameSense), the specific tool is coded instead.

Assignment rule. Does the paper evaluate a bundled RG tool suite as a whole? If yes, PROGRAM. Does it evaluate a specific tool within a programme? If yes, code the specific tool.

3.8 Low-Risk Gambling Guidelines vs Threshold-Implemented Tools

GUIDE-LRGL captures epidemiologically-derived consumption thresholds (e.g., “gamble no more than X hours per month”) that describe the scientific basis for safer-gambling guidance but are not themselves implemented as player-facing interventions. When a threshold is implemented (for example, as an automated alert triggered when the player crosses the guideline) the implementing tool is coded (typically MSG-NOTIFY, MSG-POP, or FEED-RISK as appropriate).

4. Bundled Code Handling Rules

Of the 410 extraction records, 34 (8.3%) are coded with bundled combinations (e.g., FEED-PERSONAL + MSG-POP). Bundling is used to capture tools where two or more mechanisms are integral to the design and cannot be disentangled in the study.

4.1 When to Bundle

Use a bundled code when both (or all) coded elements are integral to the tested design and removing any would change what the study tested. A paper testing whether personalised feedback delivered via an interruptive pop-up improves outcomes compared with generic pop-ups tests both the content mechanism (personalised vs generic) and the delivery mechanism (pop-up). Both elements are integral: bundle as FEED-PERSONAL + MSG-POP.

Do not bundle simply because the tool has more than one coded attribute. A personalised feedback email where the email is merely the delivery channel (not a tested mechanism) is coded FEED-PERSONAL alone. The test for bundling is substantive: would the paper’s claim differ if one element were absent? If yes, bundle; if no, do not.

4.2 Ordering of Bundled Codes

Where bundling is required, the primary mechanism (the one most central to the paper’s causal claim) is listed first. In FEED-PERSONAL + MSG-POP, FEED-PERSONAL is listed first because personalised content is the primary mechanism under test; the pop-up is the vehicle through which it is delivered. The ordering convention is used by downstream analyses, including the section-assignment rule below.

4.3 Section Assignment of Bundled Records

For analyses that require each record to be assigned to a single domain (the domain $\times$ implementation readiness table in the main manuscript reports bundled records as a separate compound group; single-domain assignment is used where an analysis needs one domain per record), bundled records are assigned to the domain of the first-listed code. Of the 34 bundled records, 14 have codes that share a section (e.g., EXCL-TEMP + EXCL-LONG, LIMIT-DEP + LIMIT-TIME) and are assigned to that shared section unambiguously. The remaining 20 span sections; first-code assignment resolves these deterministically.

Bundled codes remain visible at the tool-code level in the record-level dataset [OSF: 03_1_Taxonomy_Dataset.xlsx]; the single-domain assignment is applied only where the analytic unit requires it. Readers querying specific combinations can do so through the record-level data.

4.4 Frequency of Bundled Combinations

The most common bundled combination is FEED-PERSONAL + MSG-POP ($n = 10$), capturing personalised behavioural feedback delivered via interruptive pop-up: a design common in recent field trials of online responsible gambling tools. The second most common is EXCL-TEMP + EXCL-LONG ($n = 4$), capturing studies evaluating self-exclusion at multiple durations within the same protocol. All other bundled combinations appear three or fewer times across the 34 records. The full distribution is given in Table 02.3.2.

Table 02.3.2
Bundled Code Combinations by Frequency

Combination	n	% of bundled (34)	Section status
FEED-PERSONAL + MSG-POP	10	29.4%	Spans
EXCL-TEMP + EXCL-LONG	4	11.8%	Shared
LIMIT-DEP + LIMIT-TIME	3	8.8%	Shared
EDU-INFO + ASSESS-SELF	1	2.9%	Shared
EDU-INTERACT + FEED-NORM	1	2.9%	Spans
EDU-INTERACT + THER-GROUP	1	2.9%	Spans
EXCL-LONG + EXCL-TEMP	1	2.9%	Shared
FEED-PERSONAL + FEED-NORM	1	2.9%	Shared
FEED-PERSONAL + STRUCT-OTHER	1	2.9%	Spans
FEED-TRACK + MSG-POP	1	2.9%	Spans
LIMIT-DEP + LIMIT-BET	1	2.9%	Shared
LIMIT-DEP + LIMIT-BET + LIMIT-LOSS + LIMIT-TIME	1	2.9%	Shared
LIMIT-DEP + LIMIT-LOSS + LIMIT-TIME + LIMIT-BET	1	2.9%	Shared
LIMIT-LOSS + MSG-POP	1	2.9%	Spans
MSG-POP + EDU-VIDEO	1	2.9%	Spans
MSG-POP + FEED-PERSONAL + LIMIT-DEP	1	2.9%	Spans

Combination	n	% of bundled (34)	Section status
MSG-POP + MSG-PUSH	1	2.9%	Shared
MSG-STATIC + EDU-INFO	1	2.9%	Spans
PROGRAM + LIMIT-DEP	1	2.9%	Spans
THER-CBT + LIMIT-LOSS	1	2.9%	Spans

Note. Section status indicates whether the constituent codes share the same section (“Shared”, n = 9 combinations covering 14 records) or span sections (“Spans”, n = 11 combinations covering 20 records). Combinations are listed in descending order of frequency, then alphabetically. Where two records appear with the same constituents in different orders (EXCL-TEMP + EXCL-LONG and EXCL-LONG + EXCL-TEMP), each ordered combination is listed separately because first-code ordering is meaningful for downstream section assignment. THER-GROUP appears here in compound form only; the code is documented in the Stage 1 dataset but did not survive consolidation into the Framework, and no Stage 2 tool carries it. Active THER-* codes are defined in Section 2.5.